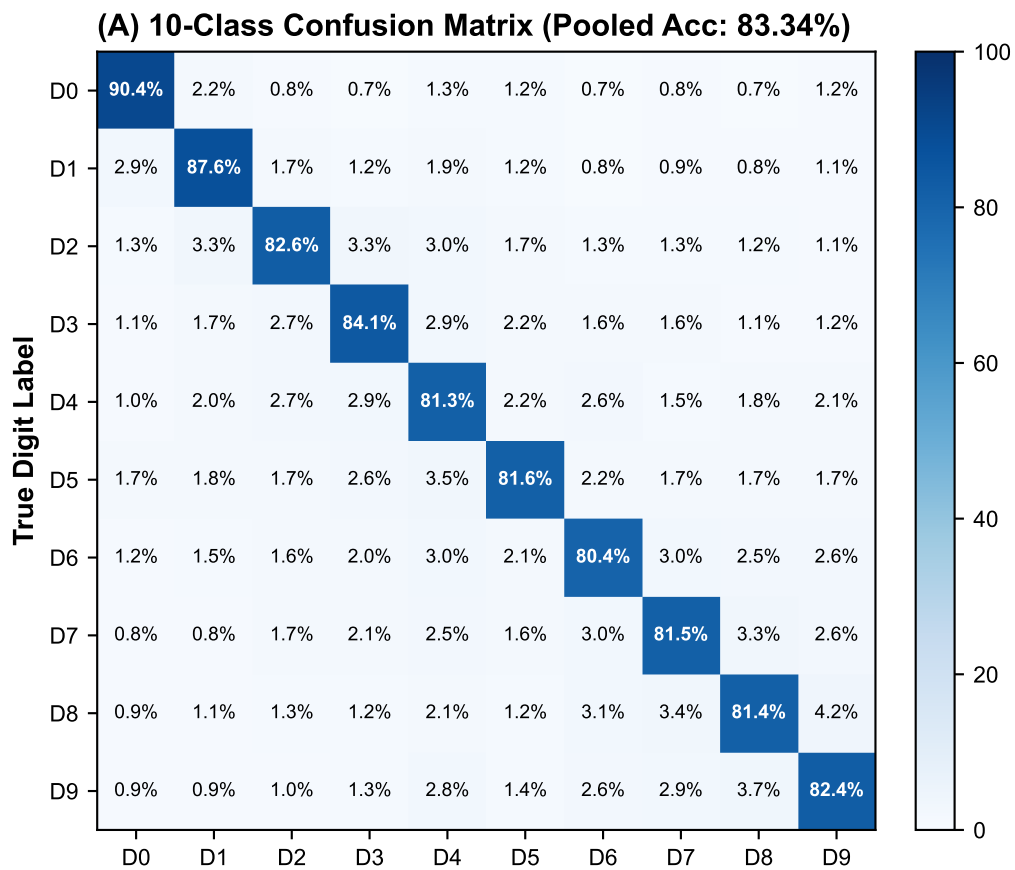
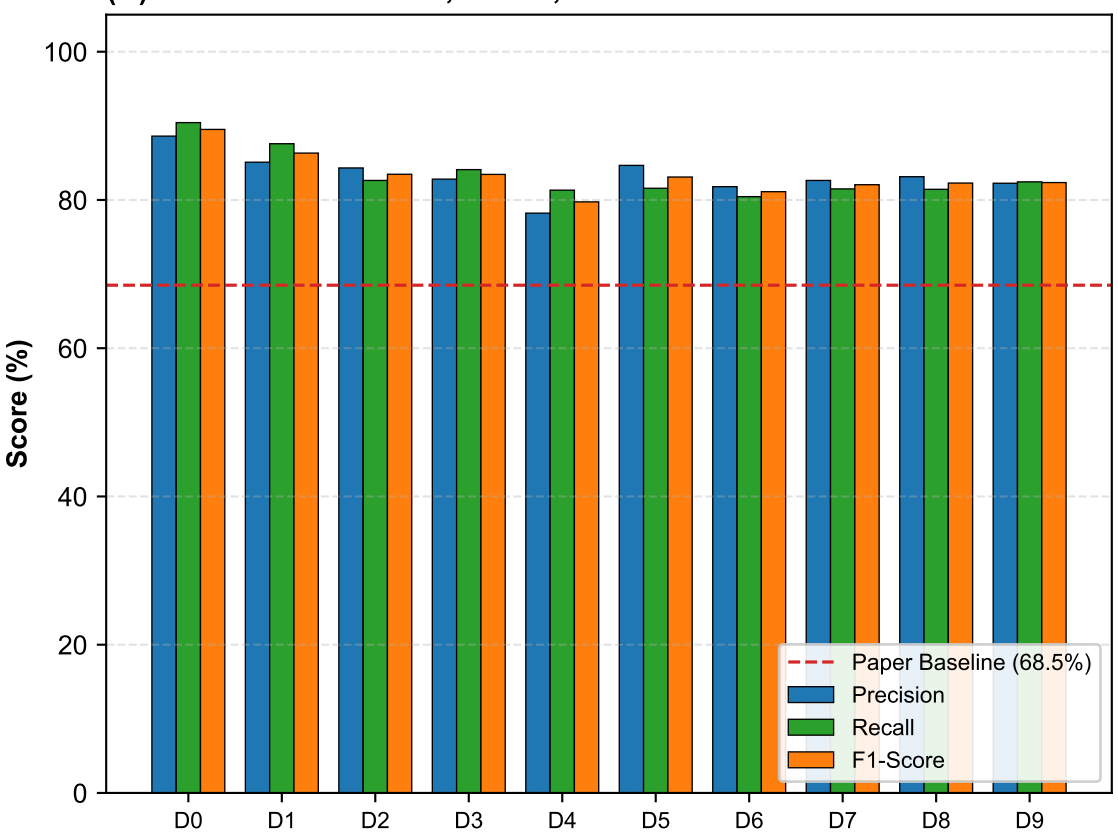
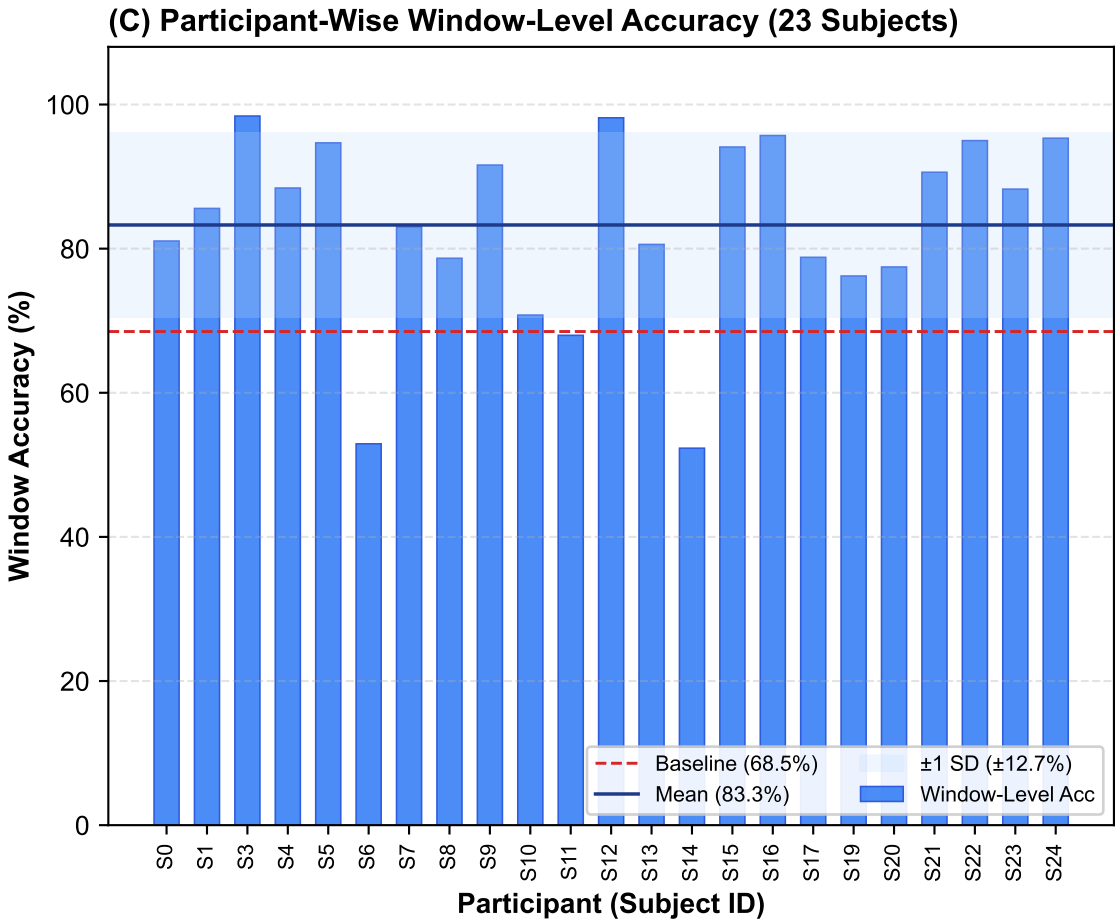
Enhanced EEG Imagined Digit Recognition — Comprehensive Experimental Evaluation







(D) Experimental Benchmark Summary

Key Evaluation Parameter	Enhanced Value	Reference / Notes
Evaluated Participants	**23 Subjects**	Subject 0 to 24
Total EEG Recordings (Trials)	**230 Files**	10 digits per subject
Total Analyzed Windows	**120,265 Windows**	50ms window, 25ms hop
Feature Vector Size	**182-D Features**	Time + Welch + Hjorth + ZC
Window Accuracy (Mean ± SD)	**83.28% ± 12.72%**	Range: 52.3% – 98.4%
Window Pooled Accuracy	**83.34%**	Δ = +14.84% over paper
Window Macro-F1 Score	**83.34%**	10-class macro average
Trial Accuracy (Mean ± SD)	**99.83% ± 0.82%**	Majority voting rule
Trial Pooled Accuracy	**99.83%**	229 of 230 trials correct
Trial Macro-F1 Score	**99.83%**	Across all 10 classes